

Total No. of Questions : 8]

SEAT No. :

PC2797

[6352]-21

[Total No. of Pages : 4

S.E. (Electronics/E & TC/Electronics & Computer Engg.)

SIGNALS & SYSTEMS

(2019 Pattern) (Semester-IV) (204191)

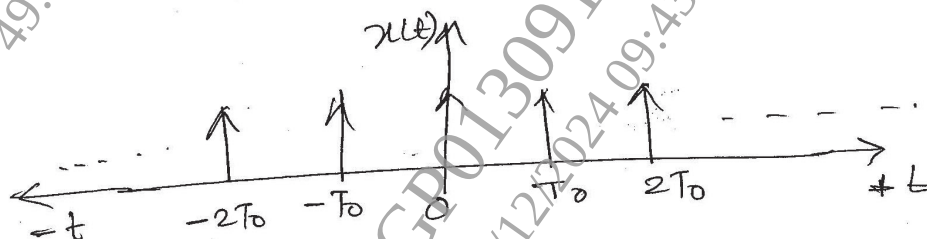
Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

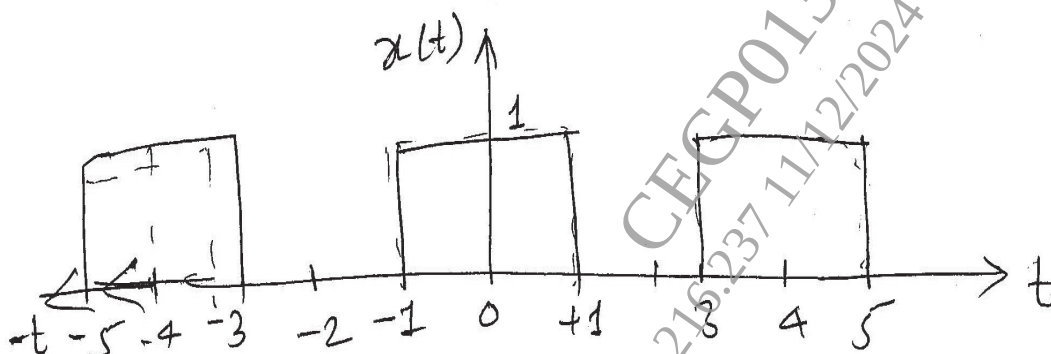
Q1) a) Find out the exponential Fourier series for impulse train shown in following figure. [8]



- b) Explain Gibb's phenomenon and how to reduce the effect. [6]
- c) Provide mathematical expression of trigonometric and exponential Fourier series. And, give the relation between their coefficients. [4]

OR

Q2) a) Find the Trigonometric Fourier series for a given signal x(t) [8]



P.T.O.

- b) State the mathematical expression and physical significance of following CT Fourier series properties. [6]

- i) Linearity
- ii) Modulation
- iii) Time shifting

- c) State the Dirichlet conditions for the existence of Fourier series. [4]

- Q3)** a) Determine the Fourier Transform of $x(t) = e^{-t} \cdot u(t)$. Also, plot magnitude and phase spectrum. [8]

- b) Find the Fourier Transform of unit step signal. [6]

- c) State and explain following properties of Fourier Transform. [3]

- i) Linearity
- ii) Time scaling

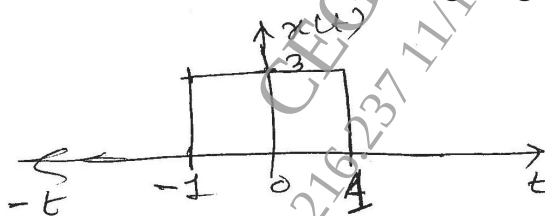
OR

- Q4)** a) Using the CT Fourier Transform properties, find the Fourier transform of [8]

i) $\frac{d}{dt} e^{-at} u(t)$

ii) $6 \text{ sinc}(0.5t)$

- b) Find the Fourier Transform of signal given below. [6]

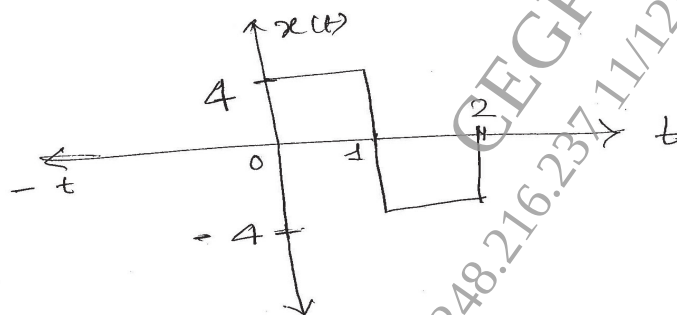


- c) Does Fourier Transform exist for the signal [3]

$$x(t) = e^{2t} \cdot u(t)$$

Justify your answer.

- Q5)** a) Determine the Laplace transform of given signal. Also, sketch ROC. [8]



- b) Determine the impulse response of the system described by the equation. [6]

$$5 \frac{d}{dt} y(t) + 10y(t) = 3x(t)$$

- c) State the conditions for stability of a system. Check whether the given [4]
 $h(t) = e^{-10t}$. $u(t)$ is stable or unstable.

OR

- Q6)** a) Find inverse laplace transform of [8]

$$X(s) = \frac{s+2}{s(s-3)(s+4)}$$

- b) Determine the initial value and final value of given [6]

$$X(s) = \frac{s+3}{s^2+2s+2}$$

- c) Determine the laplace transform of $x(t) = u(t) * u(t-3)$ [4]

- Q7)** a) CDF of a random variable is [9]

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{2x}{5} & 0 \leq x \leq 2 \\ k & x > 2 \end{cases}$$

Find:

- i) k value
 ii) $P(1 \leq x \leq 2)$
 iii) $P(x > 2)$
- b) State the properties of CDF and PDF. [8]

OR

Q8) a) PDF of a random variable is given

[9]

$$f_x(x) = \begin{cases} kx & -1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find:

- i) k ,
- ii) CDF
- iii) $P(0 < x \leq 1)$
- iv) $P(x > 1)$

b) PDF of Random variable is given as

[8]

$$f_x(x) = \begin{cases} \frac{1}{2\pi} & 0 \leq x \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

Find:

- i) Mean
- ii) Mean square value
- iii) Variance
- iv) Standard deviation

